

Q6)

Date: _____

PAC

→

Data Given

Vehicle type
Current Source of Charge
Required soc
Parking duration
Current time
Membership status
Disabled status
Station availability

→

Required Results

early exit messages
for ineligibility
• Req. charge
• Priority
• Total Discount
• Total Payable
• Duration message

→ Processing Required

- Calculate required charge amount
- Validate charging eligibility
- Assign charging priority
- Determine ^{base} charging rate based on time of day
- Calculate charging and net charging discount
- Determine base parking rate based on duration
- Accumulate discount

→ Solution alternatives

INPUT

- v Type (Vehicle type E/H)
- current SOC (current battery level %)
- Req. SOC (Required battery level %)
- duration (Expected parking duration in hours)
- Time (current time in 24-hour format)
- member (Disabled-person priority: Yes or No)
- available (charging station availability: Y/N)

Process

- Eligibility & setup
 - ↳ calculate {Req-Charge (Req. SOC - current SOC)}
 - ↳ Check if charging is available, if vehicle qualifies (Hybrid with $\geq 40\%$ SOC donot) and if charge is needed (Req-Charge > 0). Set eligible flag.
 - ↳ Determine Priority (Emergency Priority, or normal) based on, current SOC, Request SOC, disabled, and member status.

Output

- Status message
 - "Charging unavailable"
 - No charging slots available
 - Vehicle does not qualify for EV charging
 - No charging required, Final Receipt Outputs
 - Vehicle type
- Current Battery & Required Battery.
- Time states
- charging cost.

Q6) Flowchart:

Date: _____

Start

OUTPUT "Enter vehicle type (E/H): "

INPUT Vtype

OUTPUT "Enter battery charge level: "

INPUT current_soc

OUTPUT "Enter required charging level: "

INPUT Req_SOC

OUTPUT "Enter expected duration in hours: "

INPUT duration

OUTPUT "Enter current time in 24-hour format: "

INPUT Time

OUTPUT "Parking membership? (Y/N) "

INPUT member

OUTPUT "Do you have disabled person priority? "

INPUT disabled

OUTPUT "Is the charging station available? "

INPUT available

Charge_cost ← 0

Park_cost ← 0

Total_Discount ← 0

Req_Charge ← 0

Priority ← "None"

Time_Status ← "None"

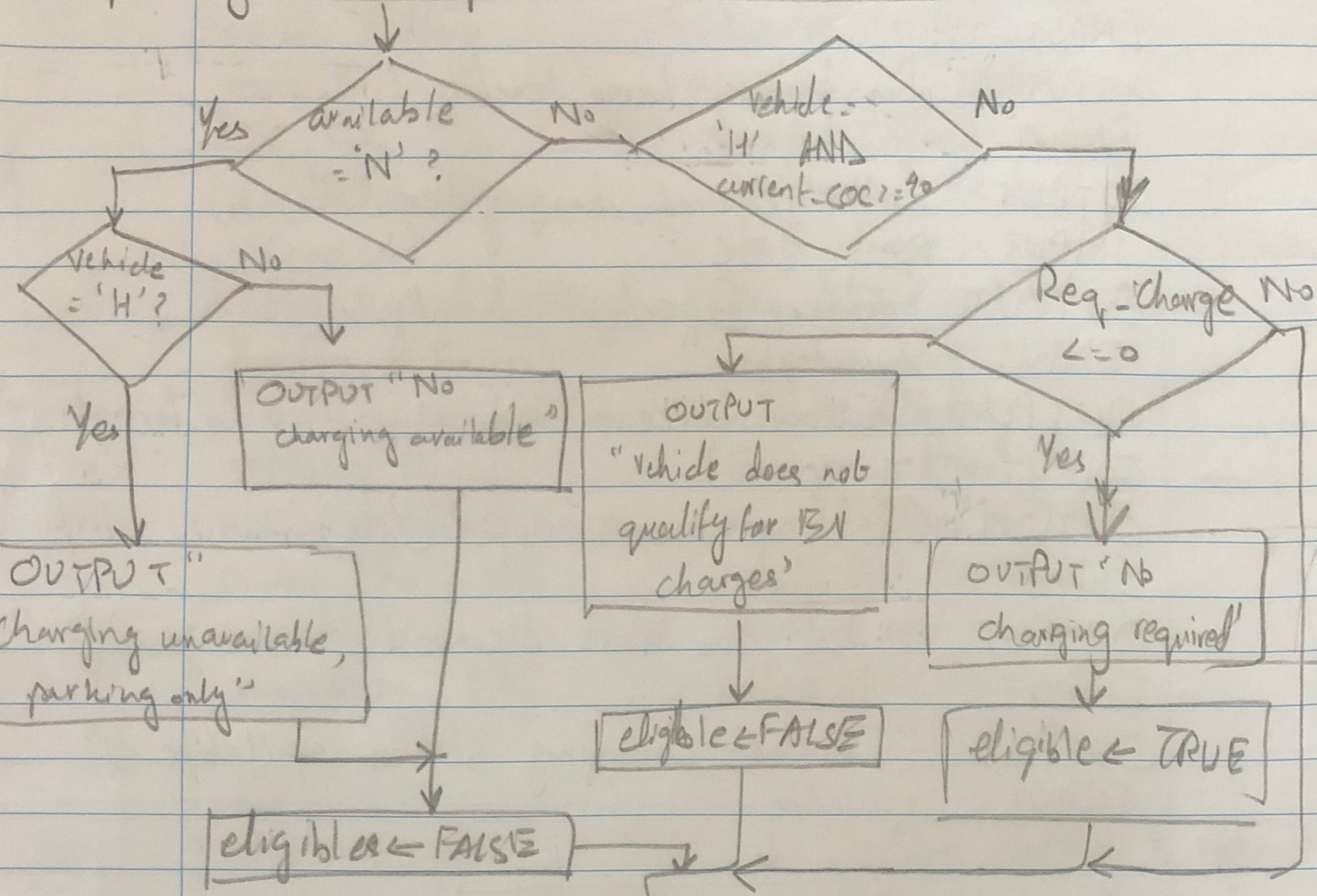


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message ← ""
eligible ← TRUE
Req_Charge ← Req_SOC - current_SOC
    
```



Priority ←
Emergency
charging
priority

